Predicting the Likelihood of Type 2 Diabetes (T2D)

Team 3c

2026.02.08

CDS-492 | Spring 2026 | Dr. Slamani

Module 01 Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza

Outline

- MD01 – First Team Meeting
  - Define roles
    - Team Lead: Tyson Johnson
    - Note taker/task scheduler: Nithila Neethi Devan
    - Assignment reviewer/submitter: Alizeh Murtaza
  - Discuss choices and ranking of the two project topics
    - Topic J : How can we use patient data to predict the likelihood of developing Type 2 Diabetes?
    - Topic H: How can a hospital optimize the allocation of its resources to improve patient outcomes?
  - Submit assignment

Meeting Minutes

Date: 01/24/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza

Agenda:

- Discuss positions/roles within the team
- Discuss data source options
- Future Assignment Planning

Notes:

- Team Selected Topic J (Type 2 Diabetes) as project focus
- Potential Data Source: South Korean Government Data Portal
  - Tyson has existing logic to translate data via Python to English.
- Literature Review Requirement: Each member should find 2-3 sources prior to next meeting; we can discuss strongest sources/what to include during the next meeting.
- Team has decided to continue with virtual meetings; next meeting is set for Jan 31st at 5pm (flexible)

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 2-3 sources for the literature review for discussion during next meeting.	Tyson, Nithila, Alizeh	01/24/2026	Started
T02					
T03					
T04					
T05					
T06					
T07					

Note:

- Status: Not started, Started, Waiting on Name of person, Date Completed
- For more tasks, add rows to the table by right clicking on a cell in the last row and choose from the horizontal menu “Insert Rows Below”
- Keep the same table for the following week(s) and add more rows

Module 02

Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza

MD02 – Revisions to work from previous module(s)

- Updated title from given question to: Predicting the likelihood of Type 2 Diabetes (Slide 1)
- Updated Task Description and Status (Slide 9)
- Minor Formatting Changed (Slide 4, Slide 5, Slide 8, Slide 9)

Meeting Minutes

Date: 02/01/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza

Agenda:

- Review first submission
- Discuss slide formatting
- Discuss literature review articles
- Complete "Data Search" slide
- Update tasks
- Schedule next meeting

Notes:

- Discussed Potential Data Sources and Factors to consider when finding data.
- Ask Professor if our question can be slightly modified to accommodate non-time-series data if longitudinal data are not publicly available.

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 1 reliable source and complete our literature review and synthesis matrix on.	Tyson, Nithila, Alizeh	01/31/2026	Completed
T02	Research Data Options	Find Data that tracks a unique identifier information, data that would allow for time series methods to be used.	Tyson, Nithila, Alizeh	01/31/2026	Started
T03					
T04					
T05					
T06					
T07					

Note:

- Status: Not started, Started, Waiting on Name of person, Date Completed
- For more tasks, add rows to the table by right clicking on a cell in the last row and choose from the horizontal menu “Insert Rows Below”
- Keep the same table for the following modules(s) and add more rows

MD02 – Literature Review | Paper 1

- Paper title: The diabetes risk score: A practical tool to predict type 2 diabetes risk
- Paper citation:
  - Lindstrom, J., & Tuomilehto, J. (2003). The Diabetes Risk Score: A practical tool to predict type 2 diabetes risk. *Diabetes Care*, *26*(3), 725–731. https://doi.org/10.2337/diacare.26.3.725
- Summary of what was investigated:
  - Study investigated whether type 2 diabetes risk can be predicted using noninvasive patient data without laboratory tests
- Results:
  - Identified key predictive variables including age, BMI, waist circumference, history of hypertension treatment, history of elevated blood glucose, physical activity, and diet.
  - The resulting Diabetes Risk Score demonstrated strong predictive performance (AUC ≈ 0.85–0.87), with scores of 9 or higher indicating substantially increased risk.
- What processing algorithms did each paper use:
  - Multivariate logistic regression, Regression coefficients converted to categorical point-based risk scoring, ROC curve analysis
- What you liked about the paper:
  - Model relies on accessible, real-world patient data, making it practical for large-scale screening.
- What additions you can provide to the topic:
  - Include additional patient data such as family history, socioeconomic factors, and race/ethnicity

MD02 – Literature Review | Paper 2

- Paper title:
  - AUSDRISK: an Australian Type 2 Diabetes Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures
- Paper citation:
  - Chen, L., Magliano, D. J., Balkau, B., Colagiuri, S., Zimmet, P. Z., Tonkin, A. M., Mitchell, P., Phillips, P. J., & Shaw, J. E. (2010). AUSDRISK: an Australian Type 2 Diabetes Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures. *Medical Journal of Australia*, *192*(4), 197–202. https://doi.org/10.5694/j.1326-5377.2010.tb03478.x
- Summary of what was investigated:
  - The development and design of AUSDRISK.
- Results:
  - AROC: 0.78 (95% CI, 0.76–0.81), HL Chi-Squared: 4.1 (P = 0.85)
  - Sensitivity: 0.74, Specificity: 0.677, PPV: 0.127
- What processing algorithms did each paper use:
  - Multiple linear regression, scoring system relying on beta scores. AROC and chi-squared for validation.
- What you liked about the paper:
  - Explained a lot of their decision-making process on things like certain features' inclusion/exclusion.
- What additions you can provide to the topic:
  - By using a different dataset and similar methodology, we can strive to reproduce similar results.

MD02 – Literature Review | Paper 3

- Paper title:
  - Opportunistic screening of type 2 diabetes with deep metric learning using electronic health records
- Paper citation:
  - Jin, Q., Zhang, H., Szczerbinski, L., Zhu, J., Gerych, W., Xu, X., Wang, K., Hsu, S., Mandla, R., Deutsch, A. J., Manning, A., Mercader, J. M., Hartvigsen, T., Udler, M. S., & Ghassemi, M. (2025). Opportunistic screening of type 2 diabetes with deep metric learning using electronic health records. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-25759-x
- Summary of what was investigated:
  - Researchers studied whether existing patient health records could be used to predict who will develop type 2 diabetes in the future using ML models.
- Results:
  - The ML model predicted Tupe 2 diabetes more accurately than traditional statistical methods. It was able to identify high risk patients earlier by detecting the various patterns that can lead to patients developing diabetes.
- What processing algorithms did each paper use:
  - Deep Metric Learning trained on Electronic Health Records
- What you liked about the paper:
  - It combines both predicting diabetes onset and identifying patient subtypes in one model, this allows for more personalized pattern detecting, and therefore interventions
- What additions you can provide to the topic:
  - The incorporation of lifestyle or social factors to improve prediction accuracy.
  - Use of explainable AI to make predictions and pattern detection process more transparent and interpretable.

MD02 – Synthesis Matrix | Paper 1

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Lindström, J., & Tuomilehto, J. (2003). The diabetes risk score: A practical tool to predict type 2 diabetes risk. Diabetes Care, 26(3), 725–731.	The paper addresses whether type 2 diabetes risk can be accurately predicted using noninvasive, easily collected patient data. The authors hypothesize that a simple risk score, based on demographic, anthropometric, behavioral, and medical history variables, can identify individuals at high risk for developing drug-treated type 2 diabetes.	Population-based cohort study using Finnish FINRISK surveys (1987 and 1992). Participants: adults aged 35–64 without antidiabetic drug treatment at baseline. Follow-up: 5–10 years using national drug registry data. Multivariate logistic regression used to identify significant predictors.	A risk score ≥9 effectively identified individuals at high risk for future drug-treated type 2 diabetes. The model demonstrated strong predictive performance (AUC ≈ 0.85–0.87). The score also performed well in identifying undiagnosed diabetes in cross-sectional analyses.	Uses noninvasive, low-cost, easily accessible patient data. Strong validation across independent cohorts. Practical and scalable for population-level screening.	Study population limited to Finnish adults, reducing generalizability. Outcome defined as drug-treated diabetes, potentially missing earlier or diet-controlled cases. Use of categorical scoring may reduce predictive precision.

MD02 – Synthesis Matrix | Paper 2

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Chen, L., Magliano, D. J., Balkau, B., Colagiuri, S., Zimmet, P. Z., Tonkin, A. M., Mitchell, P., Phillips, P. J., & Shaw, J. E. (2010). AUSDRISK: an Australian Type 2 Diabetes Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures. *Medical Journal of Australia, 192*(4), 197–202. https://doi.org/10.5694/j.1326-5377.2010.tb03478.x	"To develop and validate a diabetes risk assessment tool for Australia based on demographic, lifestyle and simple anthropometric measures." The authors explain their development of AUSDRISK.	9 risk factors categorized Pearson chi-squared test Prediction models made using multiple linear regression. Risk factors: age, sex, ethnicity, parental history of diabetes, history of high blood glucose level, use of antihypertensive medications, smoking, physical inactivity, waist circumference	AusDiab data: 1999-2000, 11,247 adults, Ages 25+, Australian 5-year follow-up: 2004-2005, 6,537/10,788 surviving returned Used 6,060 participants after removing 246 unclassifiable and 229 baseline diabetic (from AusDiab). AROC: 0.78 (95% CI, 0.76–0.81) HL Chi-Squared: 4.1 (P = 0.85) Sensitivity: 0.74 Specificity: 0.677 PPV: 0.127	The data used included a wide variety of features measured that other models may not have used, such as the use of certain medications and parental history of diabetes. The source provides a working example of a model used for predicting T2D. Can be accessed as a free "online interactive risk assessment tool." http://health.gov.au/internet/main/publishing.nsf/Content/diabetesriskassessmenttool	Acknowledges that data limitations would likely result in overly raised risks in ages under 25. May have also benefitted from a more ethnically diverse pool of participants for broad use.

MD02 – Synthesis Matrix | Paper 3

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Jin, Q., Zhang, H., Szczerbinski, L., Zhu, J., Gerych, W., Xu, X., Wang, K., Hsu, S., Mandla, R., Deutsch, A. J., Manning, A., Mercader, J. M., Hartvigsen, T., Udler, M. S., & Ghassemi, M. (2025). Opportunistic screening of type 2 diabetes with deep metric learning using electronic health records. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-25759-x	Can routine electronic health records be used to predict the future onset of type 2 diabetes and identify patient subtypes?	Trained a deep metric learning model on longitudinal EHR data to predict future Type 2 diabetes onset and learn patient patterns Used these learned patterns to cluster patients into distinct subtypes Compared model performance with logistic regression and other baseline methods	The DML model outperformed traditional prediction methods for future T2D onset prediction and revealed three distinct subtypes of patients with different clinical characteristics and responses to treatment. AUROC of *0.969* Sub types : Green (healthier, more responsive), Yellow (intermediate), Red (high comorbidity burden)	Combines disease prediction and patient subtyping within a single model Utilizes large, diverse EHR datasets Enables practical screening using existing clinical data without additional testing	Does not include lifestyle or social determinants of health Limited interpretability due to the black-box nature of deep learning

MD02 – Data Search

- Data sources
  - https://www.data.go.kr/en/data/15007122/fileData.do
- Relevance of the data to the project
  - Dataset contains features that is publicly available.
  - Dataset contains yearly health exam results of South Korean patients.
- Possible challenges with the data
  - Data is specific to the Korean demographic
  - Unsure if there is a way to incorporate timeseries methods as we are still identifying data with unique identifier information (subscriber serial number)

Module 03

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza
Tariq Abdulhak

MD03 – Revisions to work from previous module(s)

- Below is a summary of changes made on previous modules and slide numbers:
  - File name correction
  - Added Tariq Abdulhak to title slide (Slide 1)
  - “Type 2 Diabetes” to “T2D” abbreviation changes

Meeting Minutes *(use additional slides if needed)*

Date: 00/00/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza
- Tariq Abdulhak

Agenda:

- Folder Structure
- Workflow Diagram
- Business Understanding
- Data Search and Description

Notes

- Allocated Tasks per person between Workflow Diagram, Business Understanding, and Data Search post discussion
- Discussed a change in our business problem, shift the focus off of longitudinal data
- Finished on Business Understanding together based on up

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 1 reliable source and complete our literature review and synthesis matrix on.	Tyson, Nithila, Alizeh	01/31/2026	Completed
T02	Research Data Options	Find Data that tracks a unique identifier information, data that would allow for time series methods to be used.	Tyson, Nithila, Alizeh	02/05/2026	Completed
T03	Workflow Diagram	Creating the workflow diagram through Xmind	Nithila	02/05/2026	Started
T04	Black Box Diagram	Creating the back box diagram through canva	Alizeh	02/05/2026	Started
T05	Class Folder Structure	Created and Shared Class Folder	Alizeh	02/05/2026	Started
T06	Fishbone Diagram	Creating the black box diagram through canva	Alizeh	02/05/2026	Started
T07	Business Understanding	Completing the business understanding slide	Tariq, Alizeh, Tyson, Nithila	02/05/2026	Completed
T08	Data Search	Completing the Data Source Slides	Tyson	02/05/2026	Started
T09	Analytic Problem	Completing the analytic problem slide	Tariq	02/05/25	Started

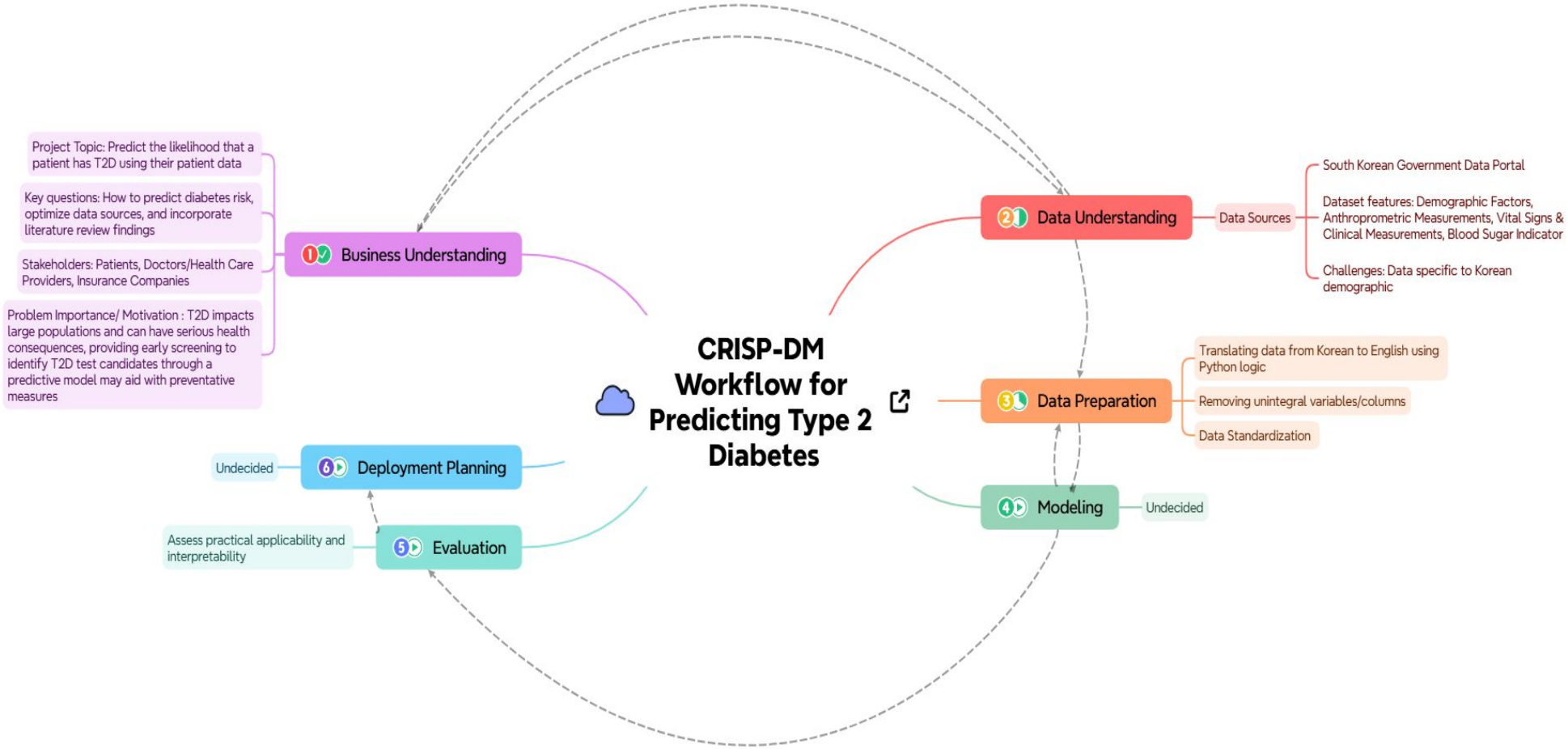
MD03 – Class Folder

- Below is a screenshot of our Team's class folder structure

```
CDS-492
 ├──── Code
 ├──── Data
|   └──── Set_01
 └──── Project
     └──── Assignments
         ├────
WK01_Team_3c_CDS-492_Project_SubmissionDate_2026-1.24.2026.pdf
         └────
WK02_Team_3c_CDS-492_Project_SubmissionDate_2026-2.01.2026.pdf
```

MD03 – Workflow Diagram

- Below is our project Workflow Diagram as of 2026.02.05



MD03 – Business Understanding

- Business problem and question(s)
  - Predict the likelihood that a patient has T2D using their patient data
- Problem importance
  - T2D is a significant problem that impacts the general population with serious health consequences
  - Diabetes is the 7th leading cause of death in the US (https://www.cdc.gov/nchs/fastats/diabetes.htm)
- Stakeholders
  - Patients
  - Doctors/Health care providers
  - Insurance companies
- Motivation
  - Provide early screening to identify type 2 diabetes test candidates
- Suggested title for the project
  - Screening T2D Using Patient Data
- Possible challenges with the problem topic
  - May not be able to create a predictive model that accurately diagnoses patients with T2D using Korean data used in this project

MD03 – Analytic Problem

- Describe the business question in terms of data, model, assumptions, limitations, and goals

Analytic Problem:

- Use patient data to predict whether a person is likely to have Type 2 Diabetes
- The goal is to help identify people who may need early screening

Data:

- Korean patient health dataset
- Includes :
  - age and gender
  - height, weight, and waist size
  - blood pressure values
  - blood sugar and cholesterol levels

Goal:

- Predict diabetes risk based on patient health measurements
- Support early detection and prevention

MD03 – Analytic Problem Continued

- Describe the business question in terms of data, model, assumptions, limitations, and goals

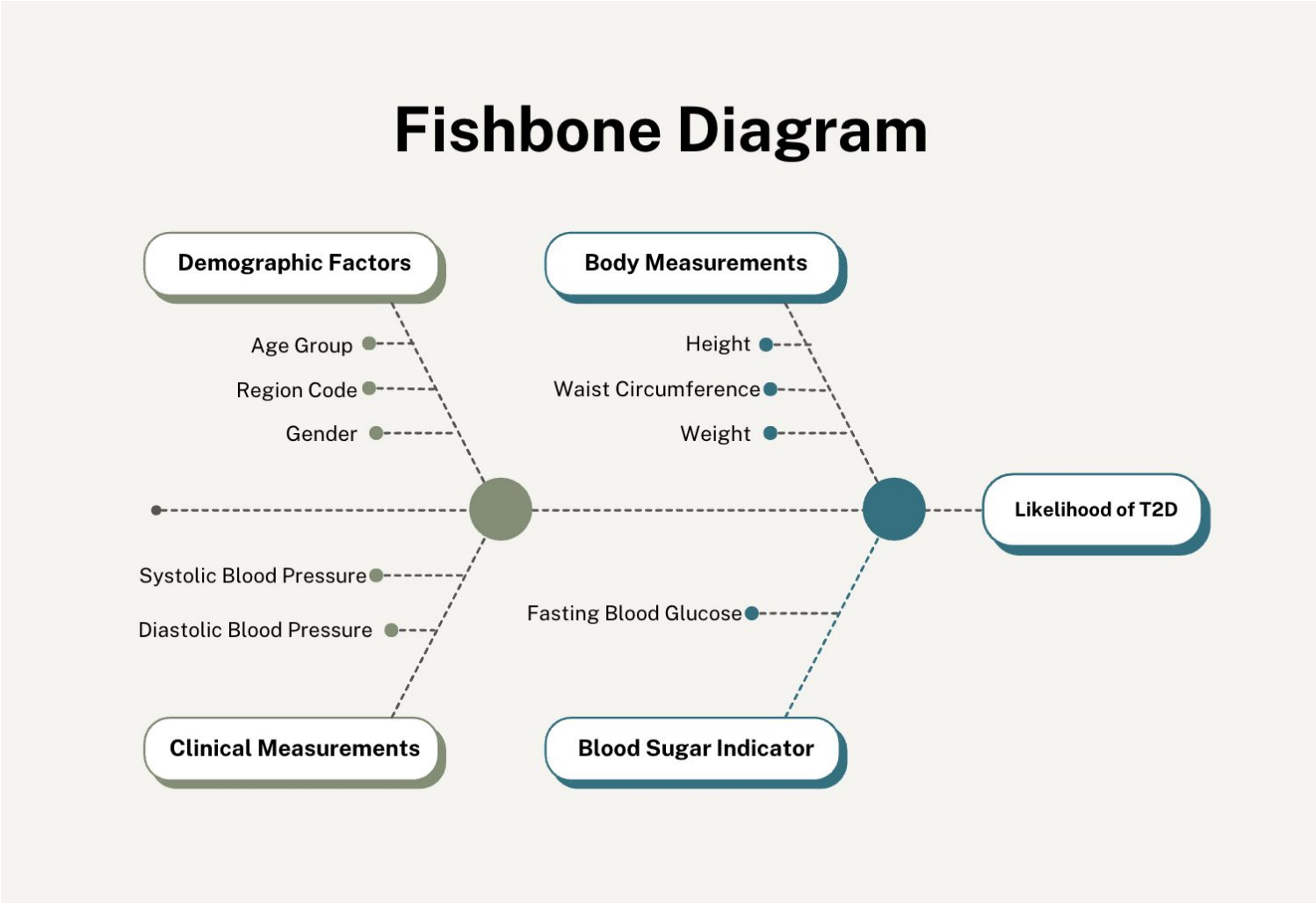
Assumptions and Limitations:

- The data is accurate and properly recorded
- Results may not apply to all populations

Important factors and categories:

- Demographics (age, gender)
- Body measurements (height, weight, waist size)
- Blood pressure
- Blood sugar levels

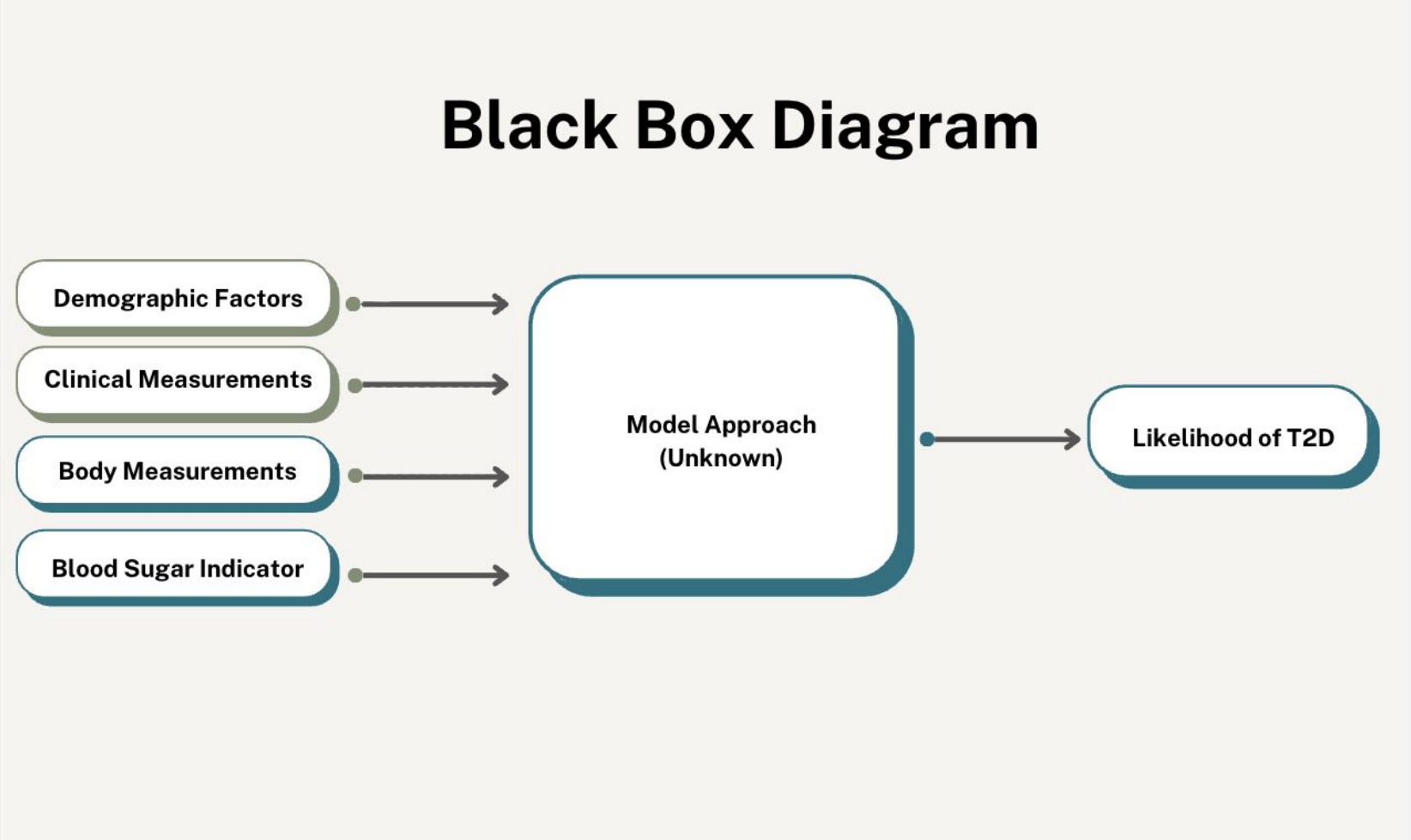
MD03 – Analytic Problem: Fishbone Diagram



MD03 – Analytic Problem: Black Box Diagram

- Inputs
  - Demographic Factors
  - Clinical Measurements
  - Body Measurements
  - Blood Sugar Indicator
- Outputs
  - Likelihood of T2D
- Modeling approach if known
  - Unknown

MD03 – Analytic Problem: Black Box Diagram



MD03 – Data Search

- Data sources
  - https://www.data.go.kr/en/data/15007122/fileData.do
  - Publicly provided by South Korea's National Health Insurance Service (NHIS)
- Relevance of the data to the project
  - Patient data of 1 million randomly selected patient exams throughout year.
  - Patient data from 2002-2024 (updated annually) available.
  - Vast majority of exams include resting blood glucose measurements.
- Possible challenges with the data
  - Some variables are rounded.
  - Data and manual recorded in Korean, requiring accurate translations.
  - Lack of identifying information prevents tracking patient over time.

MD03 – Data Description

- Entries: 1,000,000
  - Represents an insurance-covered patient aged 20+.
  - Reduced to 994,153 after removing entries missing target variable (fasting blood glucose).
- Features: 34
  - Some features are completely missing or irrelevant.
  - Basic patient details, health exam results, questionnaire information.
- CSV filetype (cp949 encoding)
- Column names are self-descriptive, but manual provides detailed explanations.
- Varying sparsity because not all patients received all tests.
  - 334,265 entries have values for all relevant variables.
- All data in this set is from 2024.

MD03 – Data Description

Feature	Minimum	Maximum	Mean	Median	St. Deviation
age_group_code*	5	18	10.64	11	3.05
height**	130	195	162.78	165	9.34
weight**	30	135	64.60	65	14.03
waist_circumference	6.8	210	81.55	81.1	10.94
systolic_blood_pressure	60	271	122.69	121	14.59
diastolic_blood_pressure	11	182	75.22	75	10.36
fasting_blood_sugar	17	991	100.98	96	23.00
total_cholesteral	25	2204	196.79	195	43.42
triglycerides	1	4622	129.32	105	99.46
hdl_cholesterol	1	908	57.09	55	15.48
ldl_cholesterol	0	2174	114.47	113	39.31
hemoglobin	0.1	25	14.22	14.2	1.53
serum_creatinine	0.06	87.2	0.84	0.8	0.37

*Grouped as 5 years (ex. 20-24). **Rounded down to nearest 5.

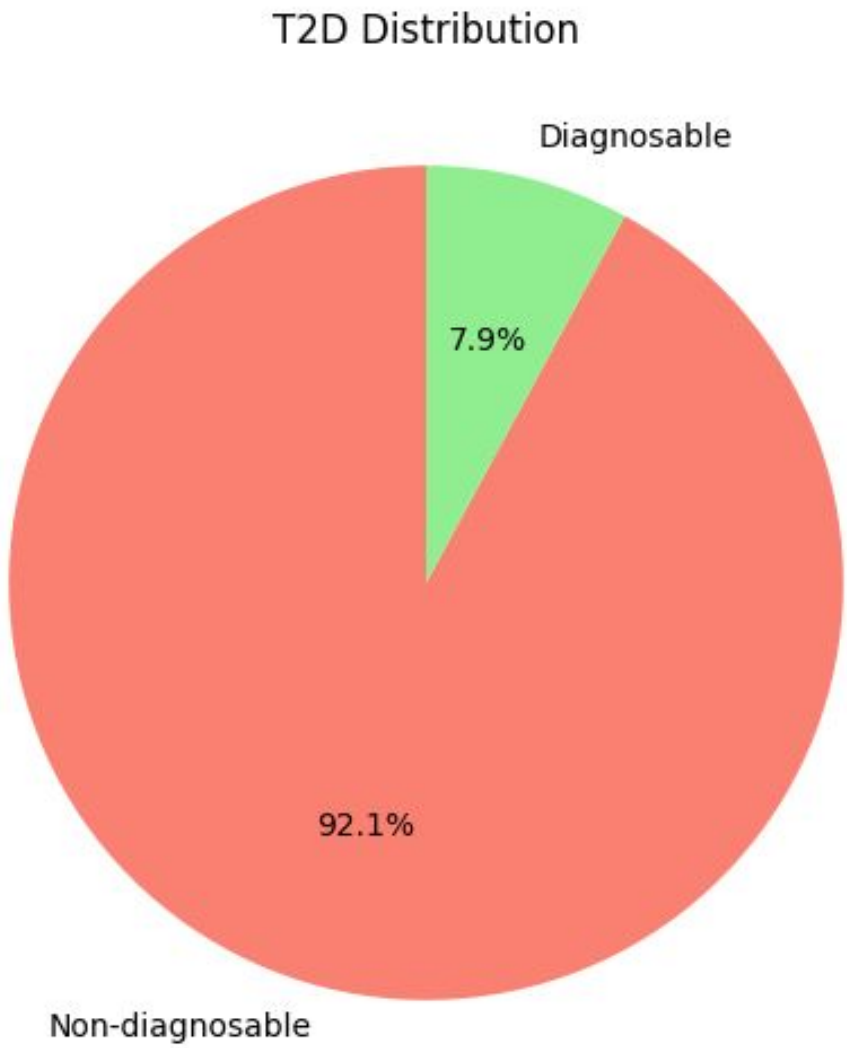
MD03 – Data Description

Feature	Minimum	Maximum	Mean	Median	St. Deviation
serum_got_ast	1	7540	26.97	23	22.644
serum_gpt_alt	1	5310	26.53	21	25.52
gamma_gtp	1	9999	34.88	22	58.61

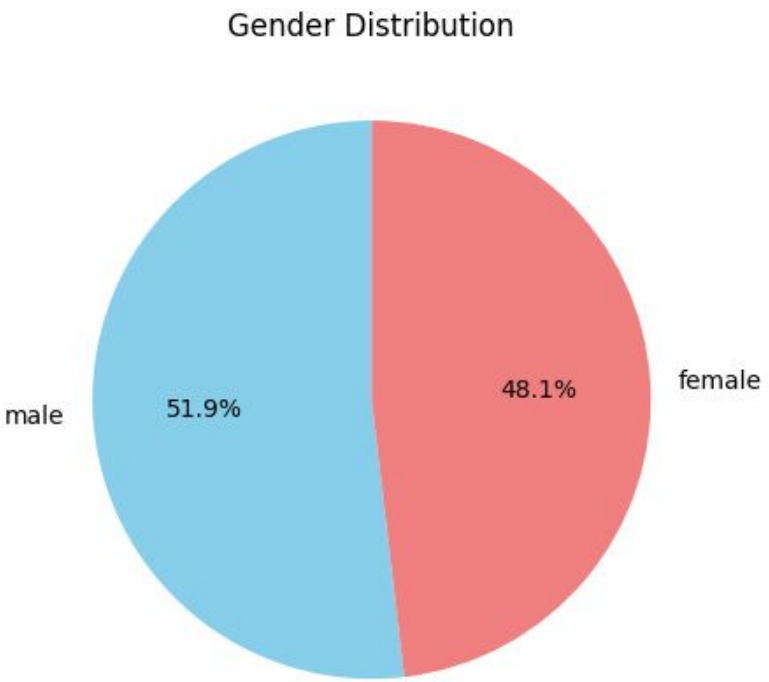
Non-numeric features recorded:

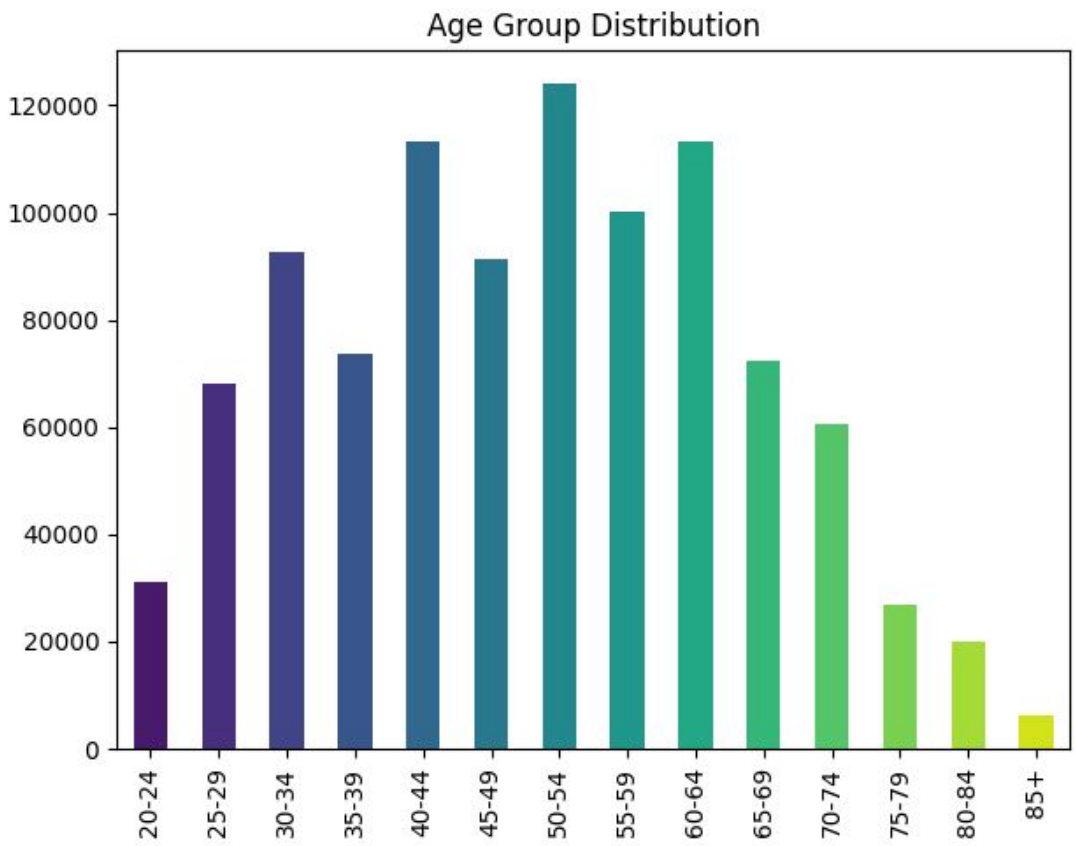
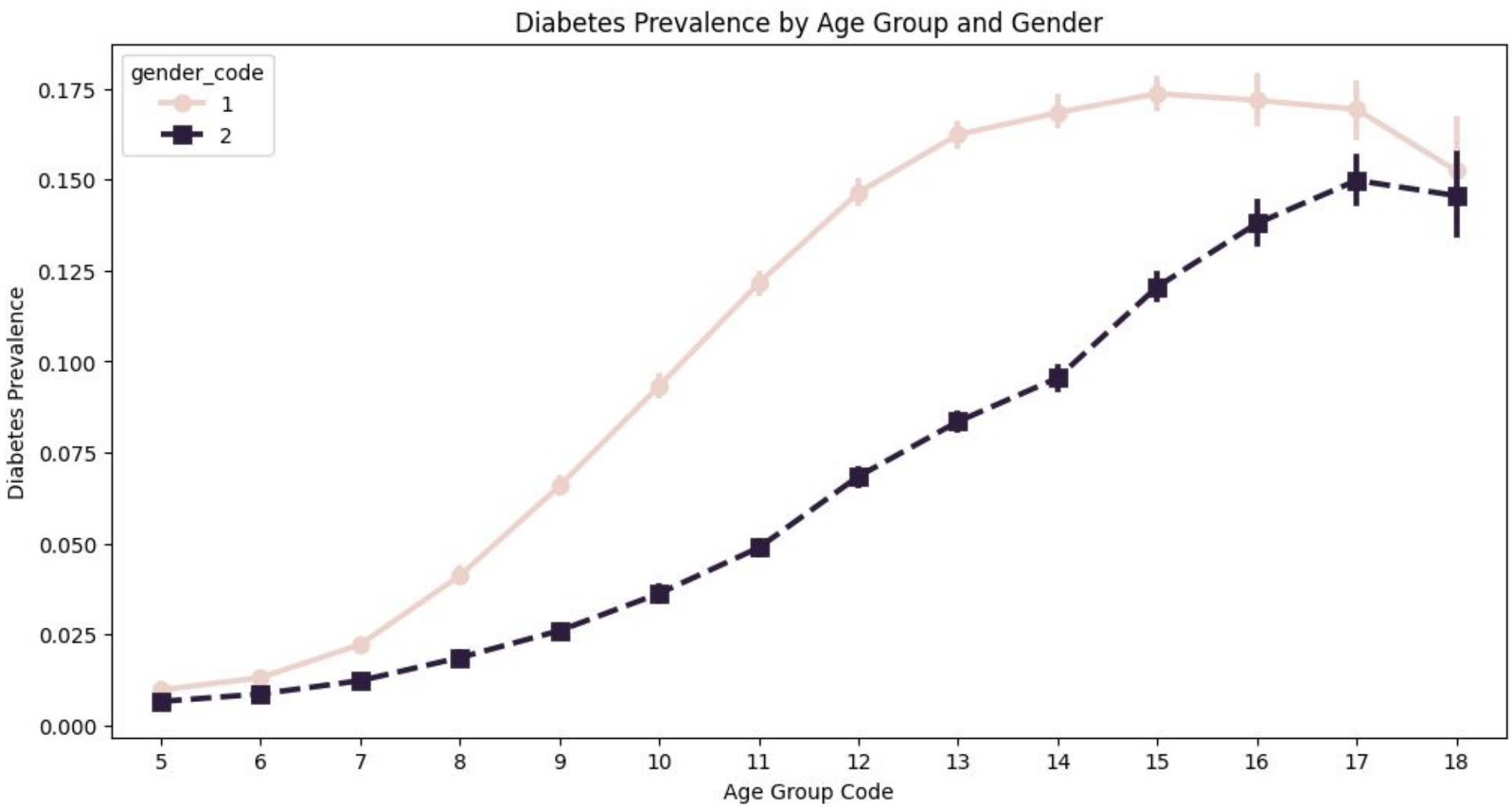
- gender_code (bool)
- vision_left/vision_right (has exception values)
- hearing_left/hearing_right (bool)
- urine_protein (categorical)
- smoking_status (categorical)
- alcohol_consumption (bool)
- oral_exam_status (bool)
- dental_caries_presence (bool)
- plaque_presence (categorical)

MD03 – Data Visualizations



MD03 – Data Visualizations





MD03 – Data Visualizations